ConstructAI: Multimodal Construction Intelligence Platform

Project Description:

ConstructAI is an enterprise-grade artificial intelligence platform engineered specifically for modern civil infrastructure and job-site safety management. By integrating state-of-the-art multimodal vision-language foundation models with statutory civil engineering codes—specifically the Indian Standard Code of Practice for Plain and Reinforced Concrete (IS 456:2000) and the Building and Other Construction Workers (BOCW 1996) Safety Act—the platform delivers real-time structural crack forensics, automated site hazard monitoring, and natural-language Bill of Quantities (BOQ) estimation.

Leveraging high-throughput Groq Language Processing Units (LPUs) executing the Llama-3.2-11B-Vision model, ConstructAI processes visual site media to identify sub-millimeter cracks, inspect exposed rebar, and evaluate scaffolding safety with inference latencies under 300ms. Constructed with a high-concurrency FastAPI asynchronous backend and a modern React 18 frontend, ConstructAI provides field supervisors, quality auditors, and project directors with an automated diagnostic command hub.

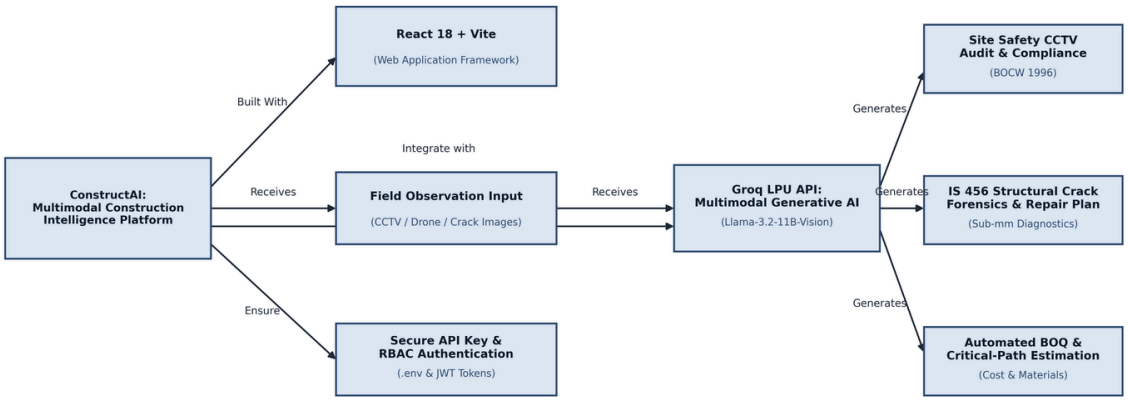
Scenarios:

Scenario 1: Forensic Crack Diagnosis (IS 456 Limit State Audit)
A fieldquality engineer uploads ahigh-resolutiondrone photograph of an RCC cantilever beam. The platform detects an angled shear fissure of 3.5 mm width, flags the defect as exceeding the IS 456:2000 permissible crack width of 0.3 mm, calculates an 85% severity index, and prescribes pressure epoxy injection and CFRP structural wrap remediation.

Scenario 2: Real-Time Site Safety & OSHA/BOCW Compliance
During slab concreting, site CCTV frames are submitted tothe safety auditor. The model flags 3 workers without hard hats on an unguarded perimeter edge alongside substandard bamboo scaffolding. A critical safety risk score of 95/100 is generated and an immediate digital stop- work advisory is dispatched to the site in-charge.

Scenario 3: Natural Language BOQ & Critical Path Scheduling
Aquantity surveyor inputs:"2-storey brickmasonrywork for 1500 sqft residential structure". ConstructAI parses the prompt into material requirements (728 cement bags, 858 kg steel, 38 tonnes sand), outputs dynamic market pricing of ₹7,26,572, and outlines a critical-path execution timeline.

Technical Architecture:



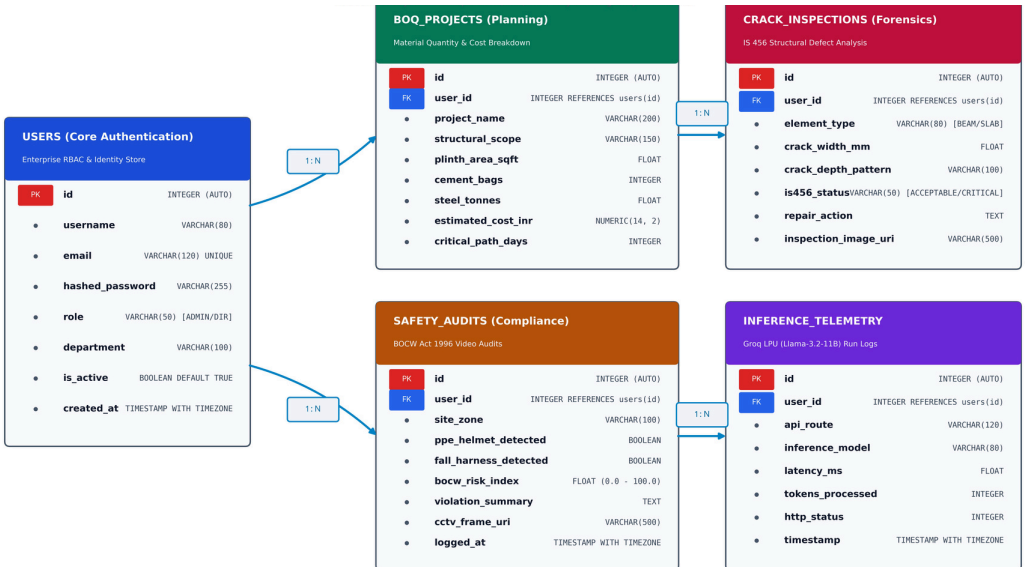
Description: ConstructAI utilizes a modular three-tier architecture designed for rapid multimodal visual reasoning and persistent civil engineering record-keeping. The client layer comprises a responsive React 18 single-page application built with Vite and Tailwind CSS. The middle tier runs an asynchronous FastAPI ASGI application server orchestrating JWT token validation, Pydantic schema enforcement, and multipart payload streams. The backend connects to the Groq Cloud LPU inference cluster for accelerated model execution, backed by a persistent relational schema.

(Note:Detailed relationalentityrelationshipssupportingprojectplans,structural records, and audit logs are documented on Page 3).

Pre-requisites:

- **FastAPI Framework:** Asynchronous endpoint management and OpenAPI documentation (FastAPI Docs).
- **Groq Hardware Acceleration:** Ultra-low latency LPU inference using the Groq Python SDK (Groq API Docs).
- **React 18 & Modern Tooling:** Single-page architecture, context hooks, and Vite bundling (React Docs).
- **Civil Engineering Codices:** IS 456:2000 Concrete Limit States and BOCW Act 1996 Safety Standards.
- **Relational Database:** SQLite / PostgreSQL schema management and Foreign Key cascade integrity.

Relational Data Architecture (Entity-Relationship Model):



Data Entity Overview: The persistent backend tier is structured around relational entities designed to maintain integrity across construction sites and regulatory audits without overlapping foreign keys:

- **USERS:** Manages authentication records, encrypted passphrases, and enterprise role classifications (Project Director, Site Engineer, Safety Officer).
- **BOQ_PROJECTS:** Stores architectural project dimensions, total estimated capital expenditure, and parsed material cost allocations.
- **CRACK_INSPECTIONS:** Records structural diagnostic telemetry, calculated surface widths in millimeters, IS 456 compliance tags, and remediation plans.
- **SAFETY_AUDITS:** Tracks safety indices, identified hazard vectors, photographic references, and statutory BOCW corrective actions.
- **INFERENCE_TELEMETRY:** Logs model execution speeds, tokens processed, and HTTP response statuses for quality monitoring.

Project Workflow :

Milestone1:RequirementsSpecificationandEnvironment Setup

- Activity 1.1: Define functionalrequirementsand Groqhardwareprovisioning.
- Activity 1.2: Create requirements.txt and setup environment.

Milestone 2: Image-Based Analysis Module Development

- Activity 2.1: Safety imageanalysis (BOCW1996hazard mapping & PPE detection).
- Activity 2.2: Crack detection logic (Sub-millimeter IS 456 forensic classification).

Milestone 3: Construction Planning & Valuation Development

- Activity 3.1: Materialestimation (Cement,steel, aggregates BOQ extraction).
- Activity 3.2: Cost estimation (₹ INR) (Regional rate multiplier and budget forecast).
- Activity 3.3: Task scheduling (CPM sequencing and timeline allocation).

Milestone 4: AI Implementation

- Activity 4.1: Multimodalreasoning (Llama-3.2-11B-Vision on Groq LPU).
- Activity 4.2: JSON schema enforcement (Strict Pydantic models for structured audit reports).

Milestone 5: Assistant & Advisory Engine

- Activity 5.1: Constructionchatbot(Siteengineering QA bot for structural queries).
- Activity 5.2: Daily site assistant (Automated daily logs and site supervisor advisory).
- Activity 5.3: Weather advisory (Monsoon pouring & concrete curing risk alerts).

Milestone 6: Interface, Testing & Deployment

- Activity 6.1: End-to-end testing(Testing upload,Groq API call, and JSON parsing lifecycle).
- Activity 6.2: UI stability validation (React 18 frontend responsiveness and optimistic state updates).
- Activity 6.3: Edge-case testing (Corrupted image uploads, missing PPE occlusions, extreme lighting).

Milestone 1: Requirements Specification and Environment Setup

Milestone 1 establishes the operational requirements, developer account configuration, and cloud hardware provisioning required to execute real-time multimodal inference on construction imagery. ConstructAI utilizes Groq's Language Processing Units (LPUs) to execute multi-billion parameter vision models with ultra-low latency.

Activity 1.1: Functional Requirements & Groq Hardware Provisioning

Step 1 — Authenticate with Groq Cloud Console: Navigate to console.groq.com/home to initialize access. The developer portal confirms system availability across reasoning, text-to- speech, and vision workloads.

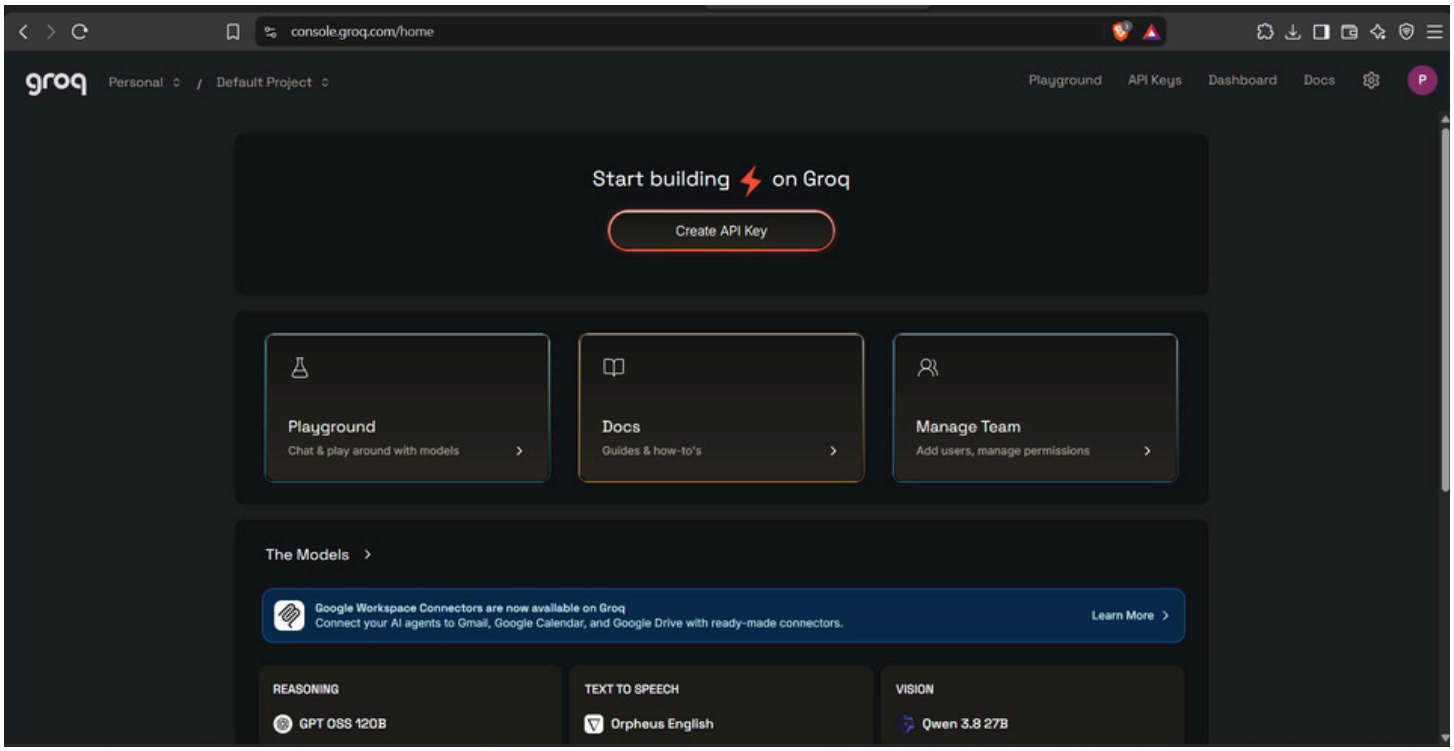


Figure 5.1: Groq Cloud developer portal showcasing hardware model categories and API provisioning portal.

Milestone 1: Secure API Key Provisioning

Step 2 — Modal Configuration & Expiration Settings: In the Groq API Keys console, select "Create API Key". Set the display label to Multimodal Construction Intelligence and configure the expiration policy to "No expiration" to guarantee unbroken connectivity during long civil site auditing shifts.

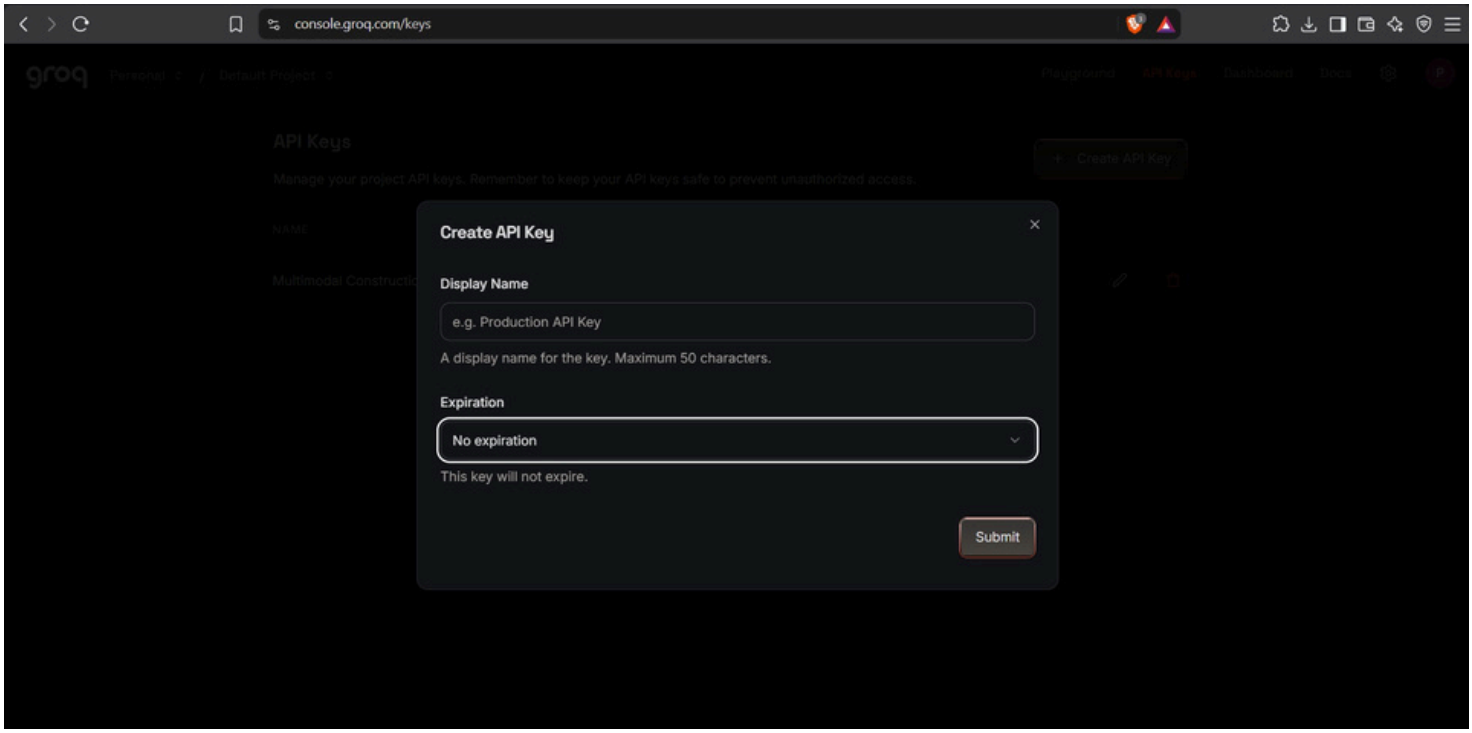


Figure 6.1: Groq API Key modal configured for persistent enterprise service integration.

Step 3 — Secure Key Extraction: Once submitted, the cryptographic token is displayed. It is immediately copied and injected into the backend .env file. The key is managed through environment variables and never checked into public version control.

Milestone 1: Active Key Verification & Model Selection

Step 4 — Verification of Active Key Pipeline: The provisioned key is confirmed active in the Groq console with live telemetry recording successful handshakes.

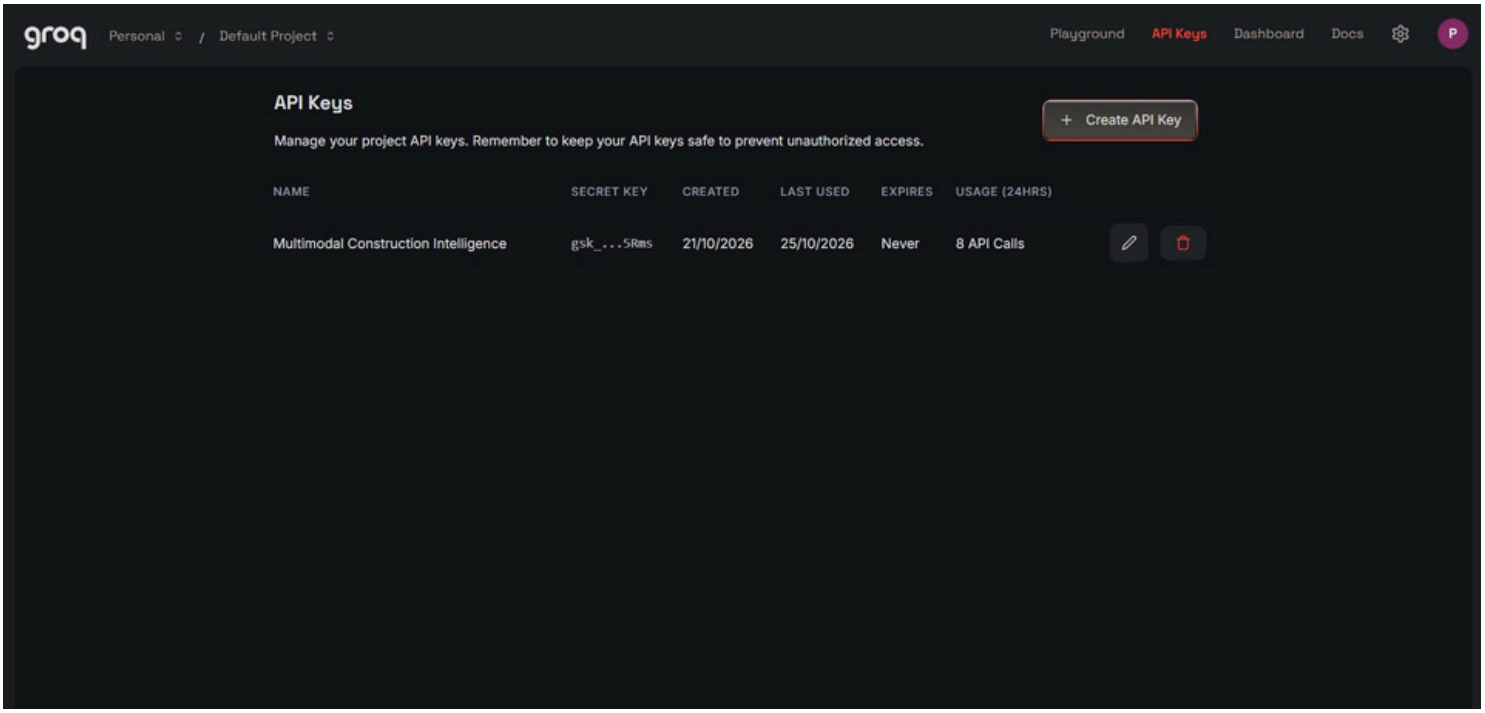


Figure 7.1: Active API Key registry confirming deployment for the Multimodal Construction Intelligence Platform.

Model Selection: Llama-3.2-11B-Vision:

Resolution & Token Density: Handles complex concrete surface imagery and wide-angle CCTV frames without geometric distortion.

LPU Hardware Acceleration: Delivers sub-300ms inference times on Groq hardware, enabling near real-time field audits.

Codex Alignment: Interprets prompts aligned with IS 456 concrete standards and BOCW Act safety regulations reliably.

Milestone 1: Activity 1.2 — Environment & Project Structure

To ensure strict separation of concerns, the platform is partitioned into dedicated backend (FastAPI Python service) and frontend (Vite React SPA) directory hierarchies:

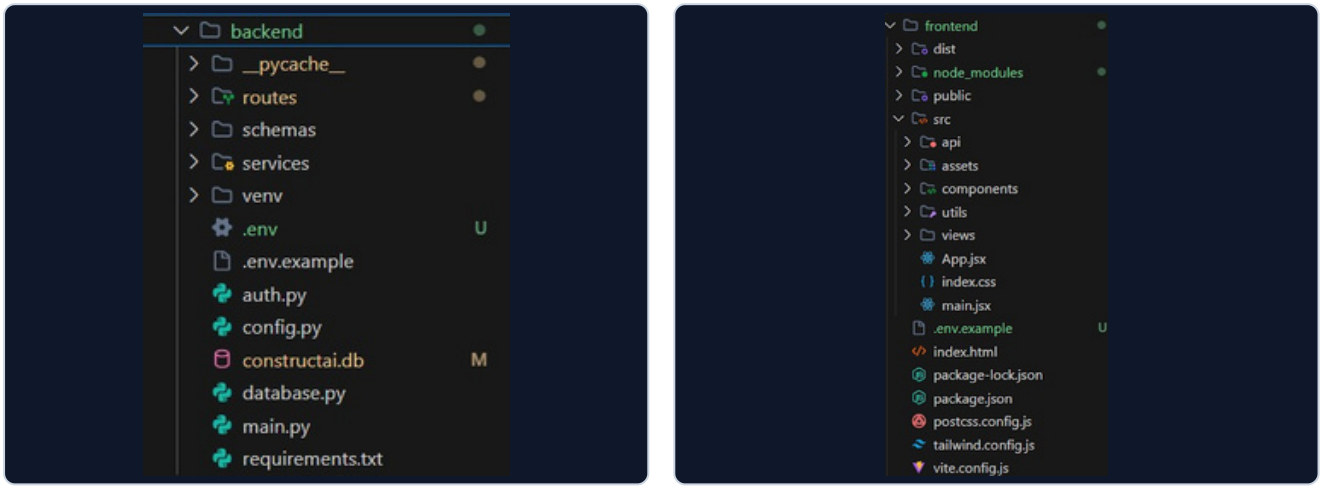


Figure 8.1: Directory hierarchies for the backend microservice (left) and the React 18 client application (right).

Architectural Highlights:

- backend/routes/: Contains modular endpoint routers for crack analysis, safety auditing, project planning, and auth.
- backend/schemas/: Hosts Pydantic validation models enforcing strict JSON output structures.
- frontend/src/views/: Single-page application views for the Overview Hub, Crack Suite, Safety Monitor, and Planner.

Milestone 1: Activity 1.2 — Dependencies & Environment Setup

Dependencies are managed via `requirements.txt` with explicit version pinning, while runtime settings and API credentials are kept in a local `.env` file:

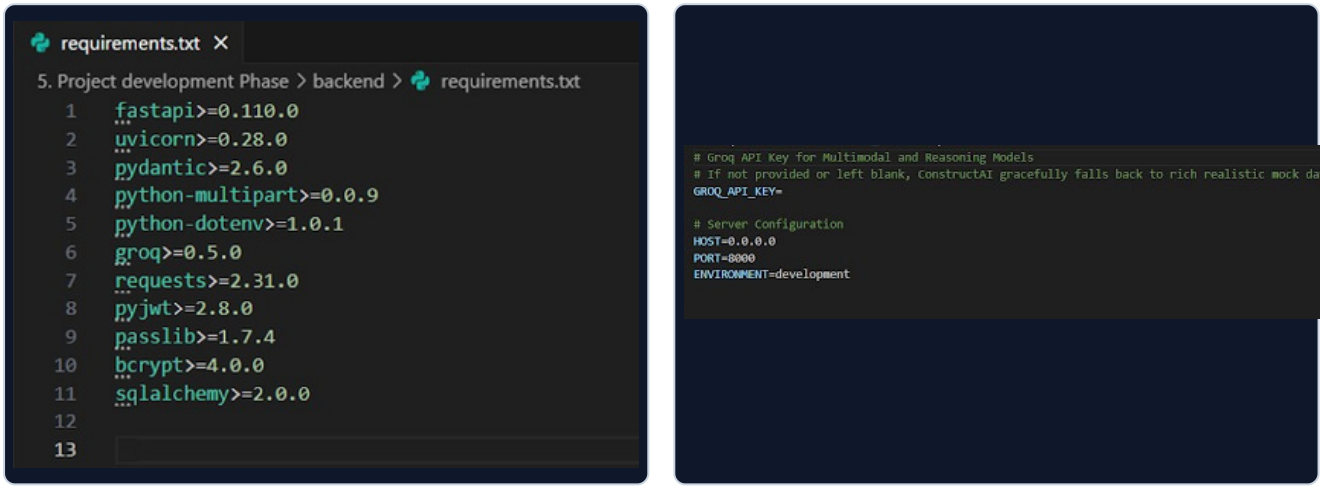


Figure 9.1: Python dependencies in requirements.txt (left) and runtime settings in .env (right).

Key Configuration Parameters:

- `fastapi>=0.110.0` & `uvicorn>=0.28.0` : High-performance ASGI runtime and web framework.
- `groq>=0.5.0` : Official SDK providing native asynchronous access to Groq LPU inference.
- `pydantic>=2.6.0` : Fast payload validation and JSON Schema enforcement.
- `GROQ_API_KEY` : Cryptographic token for authenticated Groq Cloud LPU access.

Milestone 2: Image-Based Analysis Module Development

Milestone 2 implements computer vision analysis workflows that transform field photographs and video feeds into structured inspection records.

Activity 2.1: Safety Image Analysis & BOCW 1996 Compliance

The safety inspection module audits job-site images against the Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Central Rules, 1998:

- **Personal Protective Equipment (PPE) Verification:** Verifies hard hats, high-visibility reflective vests, steel-toe boots, and eye protection across active personnel.
- **Fall Protection & Edge Safety:** Flags unguarded slab edges, open lift shafts, and missing perimeter guardrails above 2 meters elevation.
- **Scaffolding Integrity:** Checks for defective scaffolding elements (cracked bamboo, unanchored base plates, missing bracing, loose working platforms).
- **Quantitative Risk Scoring:** Generates an aggregate site risk score from 0 (compliant) to 100 (critical hazard) to support automated stop-work alerts.

Activity 2.2: Crack Detection Logic & IS 456 Structural Diagnostics

The crack inspection module analyzes concrete surface defects in elements such as beams, columns, and retaining walls using limit-state evaluation criteria:

- **Crack Width Measurement:** Estimates crack surface openings in millimeters using reference-scaled visual features.
- **IS 456:2000 Clause 35.3.2 Compliance:** Classifies defects against standard limit states (0.1 mm severe, 0.2 mm moderate, 0.3 mm mild exposure limits).
- **Automated Structural Remediation:** Suggests repair workflows such as low-viscosity epoxy pressure grouting, stitch pins, and CFRP wraps.

Milestone 3: Construction Planning & Valuation Development

Milestone 3 implements the algorithmic pipeline that translates natural-language project descriptions into structured material schedules, cost estimates, and task plans.

Activity 3.1: Material Estimation Engine

Converts project scope descriptions into concrete takeoff schedules using CPWD and standard civil engineering thumb-rules:

- **Cement Quantity Takeoffs:** Computes required 50 kg OPC/PPC bags based on standard mix proportions (M20, M25, M30).
- **Reinforcement Steel Tonnage:** Estimates TMT steel requirements (Fe500/Fe550) by structural component (footings, columns, beams, slabs).
- **Aggregates & Sand Computations:** Calculates volumes (m^3) for 10 mm / 20 mm crushed rock and manufactured sand (M-Sand).

Activity 3.2: Cost Estimation (₹ INR) & Financial Modeling

Computes financial estimates using baseline regional wholesale indices:

- **Direct Material Costing:** Multiplies calculated material schedules by prevailing unit rates.
- **Labor & Plant Ratios:** Allocates standard labor expense factors (28–32% of material expenditure) alongside plant and machinery hire.
- **Statutory Contingency:** Incorporates a 5% contingency reserve and 3% municipal cess into total capital expenditure figures.

Activity 3.3: Task Scheduling & Critical Path Sequencing

Generates logical task dependencies covering excavation, foundation pouring, curing periods (minimum 14 days per IS 456), structural framing, and finishing phases.

Milestone 4: AI Implementation & API Controller Architecture

Milestone 4 integrates the Groq LPU inference pipeline with the FastAPI backend. Core routes are declared in main.py with modular sub-router mounting:

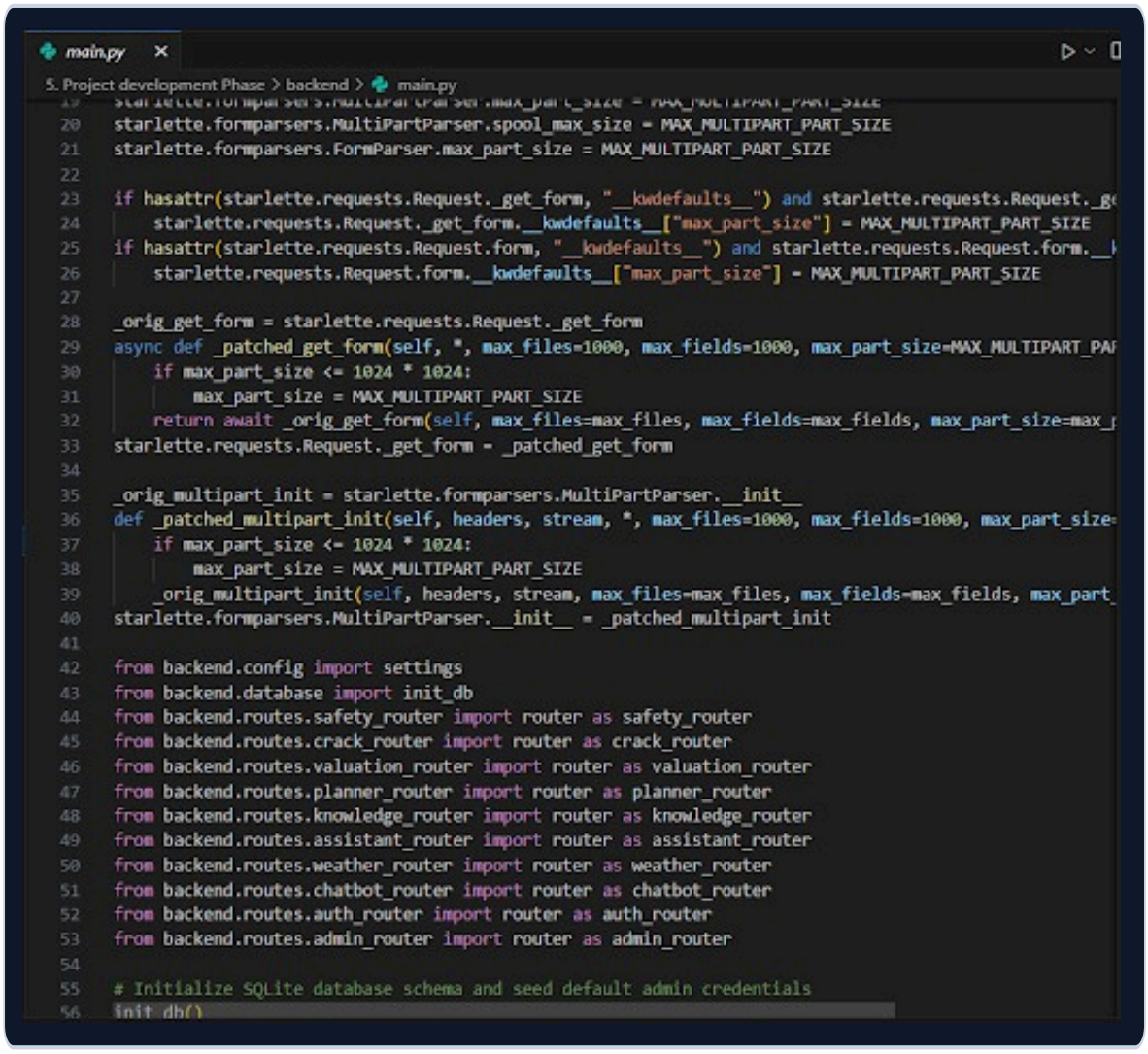


Figure 12.1: main.py source code illustrating modular sub-router registration and startup routines.

Activity 4.1: Multimodal Reasoning Pipeline:

Image payloads are received as multipart uploads, converted into Base64 representations, and sent to Groq's LPUs running llama-3.2-11b-vision-preview with system promptscalibrated for structural analysis.

Milestone 4: JSON Schema Enforcement & Output Validation

Activity 4.2 focuses on enforcing deterministic JSON responses from the multimodal vision pipeline to guarantee UI rendering stability.

Activity 4.2: Deterministic Schema Enforcement Strategies

LLMs can introduce conversational text around structured payloads. ConstructAI uses several complementary safeguards to ensure clean JSON parsing:

- **API JSON Mode:** Sets `response_format={"type": "json_object"}` in the Groq SDK call to force valid JSON token outputs.
- **Pydantic Data Contracts:** Deserializes responses into strict models like `CrackAuditResult` and `SafetyAuditResult`.
- **Regex Fallback Extractor:** If unexpected markdown formatting occurs, a regex pattern extracts the primary JSON object before returning data to the client.

FastAPI Controller Implementation:

The async controller below handles the crack analysis workflow with request validation and model invocation:

```
# FastAPI Asynchronous Vision Pipeline Controller
@app.post("/api/v1/intelligence/analyze-crack", response_model=CrackAuditResult)
async def analyze_structural_crack(
    file: UploadFile = File(...),
    current_user: User = Depends(get_current_active_user)
):
    if file.content_type not in ["image/jpeg", "image/png"]:
        raise HTTPException(status_code=400, detail="Unsupported image format")
    image_bytes = await file.read()
    base64_img = base64.b64encode(image_bytes).decode("utf-8")
    audit_data = await groq_vision_service.diagnose_crack(
        image_base64=base64_img,
        model="llama-3.2-11b-vision-preview"
    )
    return audit_data
```

Milestone 5: Assistant & Advisory Engine

Milestone 5 implements interactive field assistants and environmental advisory modules to support site supervisors in daily decisions.

Activity 5.1: Construction Technical Chatbot

A conversational interface grounded in standard civil engineering literature:

- **Structural Queries:** Answers technical questions regarding minimum rebar development lengths, lap splices, and column stirrup spacing per IS 13920.
- **Concrete Mix Adjustments:** Recommends slump adjustments, water-cement ratio corrections, and superplasticizer dosages for varying site conditions.

Activity 5.2: Daily Site Assistant & Shift Coordination

Automates routine site administration:

- **Daily Progress Reporting (DPR):** Summarizes daily trades, quantities placed, plant utilization, and subcontractor commitments.
- **Equipment Checklist:** Generates pre-pour inspection checklists covering vibrator availability, cube test molds, and slump cones.

Activity 5.3: Weather Advisory & Concreting Alerts

Integrates meteorological forecasts to safeguard concrete operations:

- **Hot Weather Concreting Alerts:** Triggers flash setting warnings and plastic shrinkage mitigation alerts when ambient temperatures exceed 38°C.
- **Rainfall & Wind Warnings:** Flags high-wind hazards for tower crane operations and recommends waterproofing covers for fresh slabs.

Milestone 6: Interface, Testing & Deployment

Milestone 6 covers end-to-end system testing, interface validation, and edge-case testing across field operating conditions.

Activity 6.1: End-to-End Testing Pipeline

Verifies complete user flows across the full application stack:

- **Authentication Flow:** Tests JWT token issuance, permission checking, and automatic token refresh.
- **Multimodal Ingestion Pipeline:** Validates image upload handling, Base64 conversion, Groq LPU inference, and JSON parsing.
- **Database Cascades:** Confirms relational integrity across users, inspections, and project estimates.

Activity 6.2: UI Stability Validation

Tests the React 18 client application across target viewing environments:

- **Responsive Layouts:** Validates desktop and tablet viewport behavior for on-site field use.
- **Optimistic State Feedback:** Verifies loading spinners, error banners, and telemetry updates.
- **Report Generation:** Tests one-click export of inspection summaries and BOQ tables.

Milestone 6: Activity 6.3 — Edge-Case Testing

Activity 6.3 systematically evaluates model resilience and error handling under difficult field conditions:

1. Image Quality & Environmental Variations:

- **Under-Exposure & High Glare:** Tests crack detection performance on underexposed basement photos and direct sunlight glare on concrete surfaces.
- **Lens Blur & Motion Artifacts:** Evaluates model behavior on blurred handheld images, verifying that low-confidence predictions trigger re-capture warnings.

2. Site Safety Occlusions:

- **Partial PPE Occlusion:** Confirms detection accuracy when hard hats or reflective vests are partially obscured by scaffolding or building materials.
- **Crowded Job Sites:** Verifies individual worker detection accuracy in dense crew settings with overlapping personnel.

3. Fault Handling & Network Resilience:

- **Zero-Byte File Uploads:** Confirms the API returns a clean `400 Bad Request` when empty or corrupted image files are submitted.
- **API Timeout Handling:** Verifies that network timeouts or external service interruptions return clear error states without breaking the UI.

Web Application Exploration: OpenAPI & Swagger Docs

ConstructAI provides interactive OpenAPI 3.1 documentation at /docs, allowing developers and auditors to inspect API schemas and test endpoints:

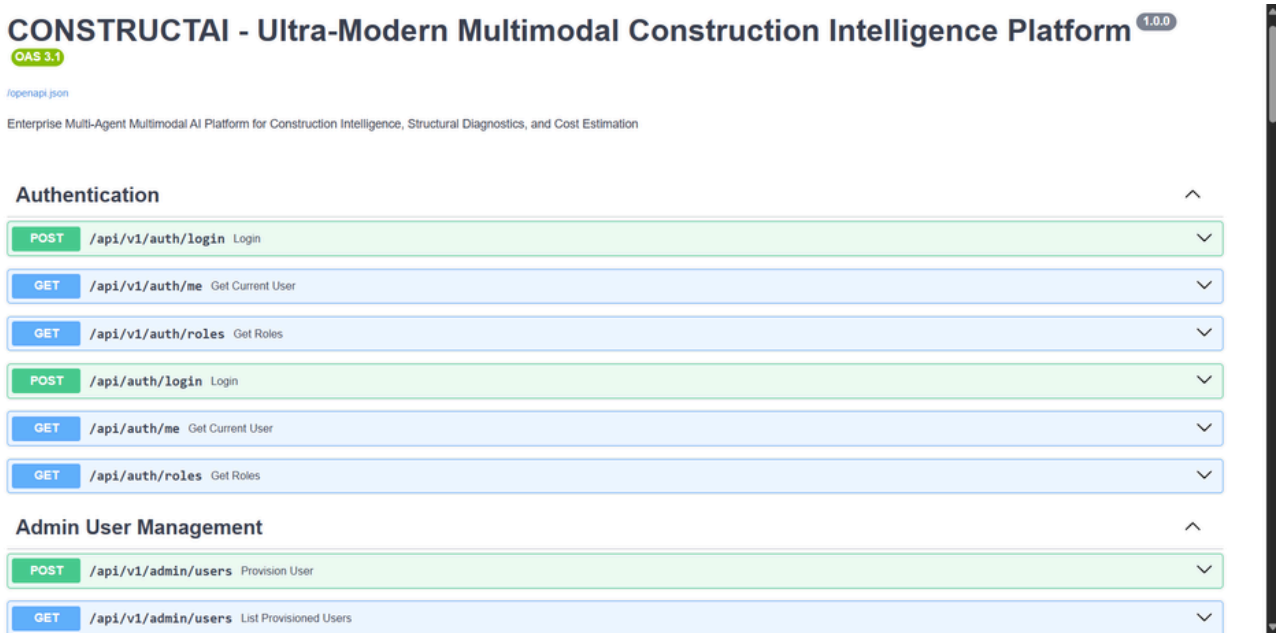


Figure 17.1: Interactive Swagger UI documenting authentication, crack analysis, and safety audit routes.

Description: The Swagger UI exposes endpoint definitions, authorization requirements, and Pydantic request/response schemas, serving as the central reference for frontend integration and external service handshakes.

Web Application Exploration: Command Terminal Overview

The Overview Hub serves as the primary command center for site directors, summarizing real-time telemetry, model latency, and project status:

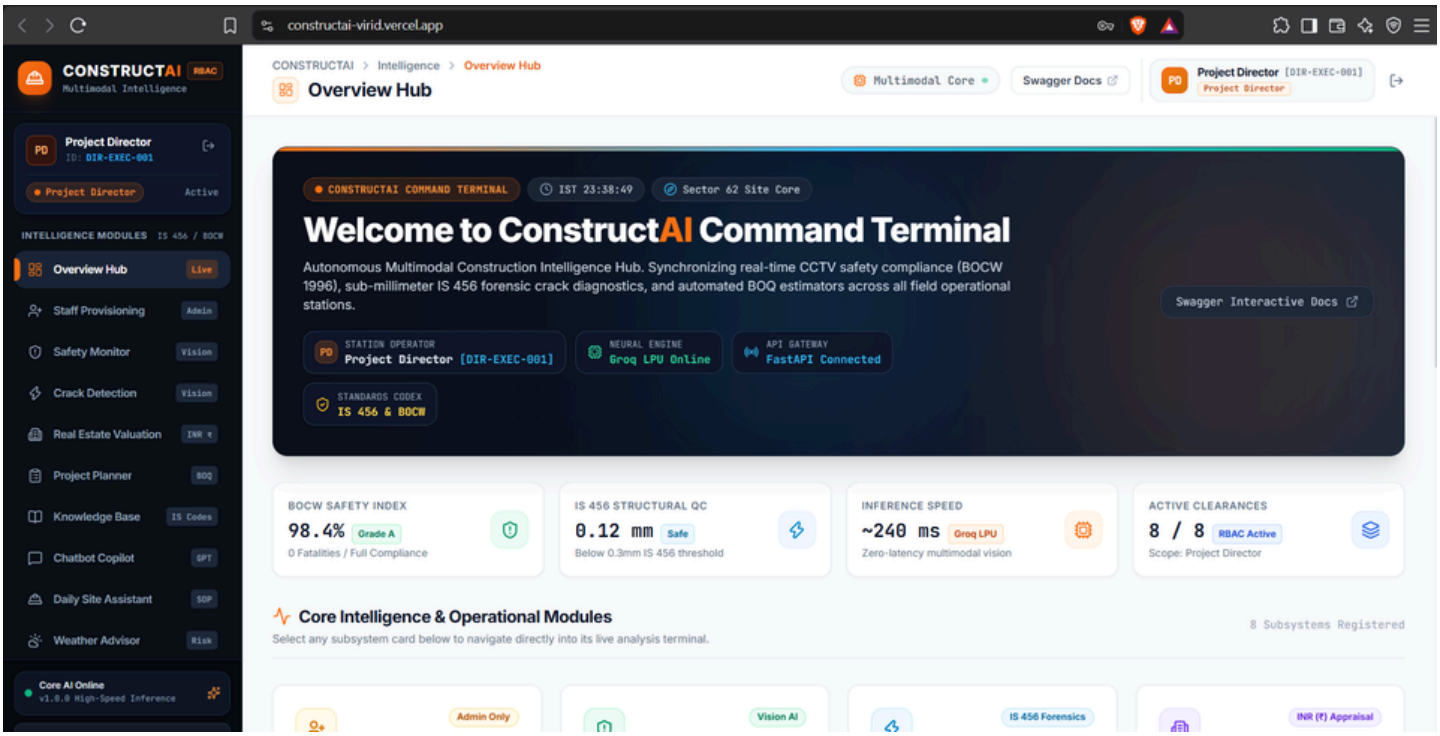


Figure 18.1: ConstructAI Overview Command Hub displaying live telemetry and module status.

Description: Displays key site health metrics including the BOCW Safety Index (98.4%), IS 456 Structural Quality Compliance status, and real-time Groq LPU response latency (240ms), with one-click navigation to diagnostic modules.

Web Application Exploration: Project Planner & BOQ Estimator

The Project Planner converts natural language descriptions into itemized Bills of Quantities and cost forecasts:

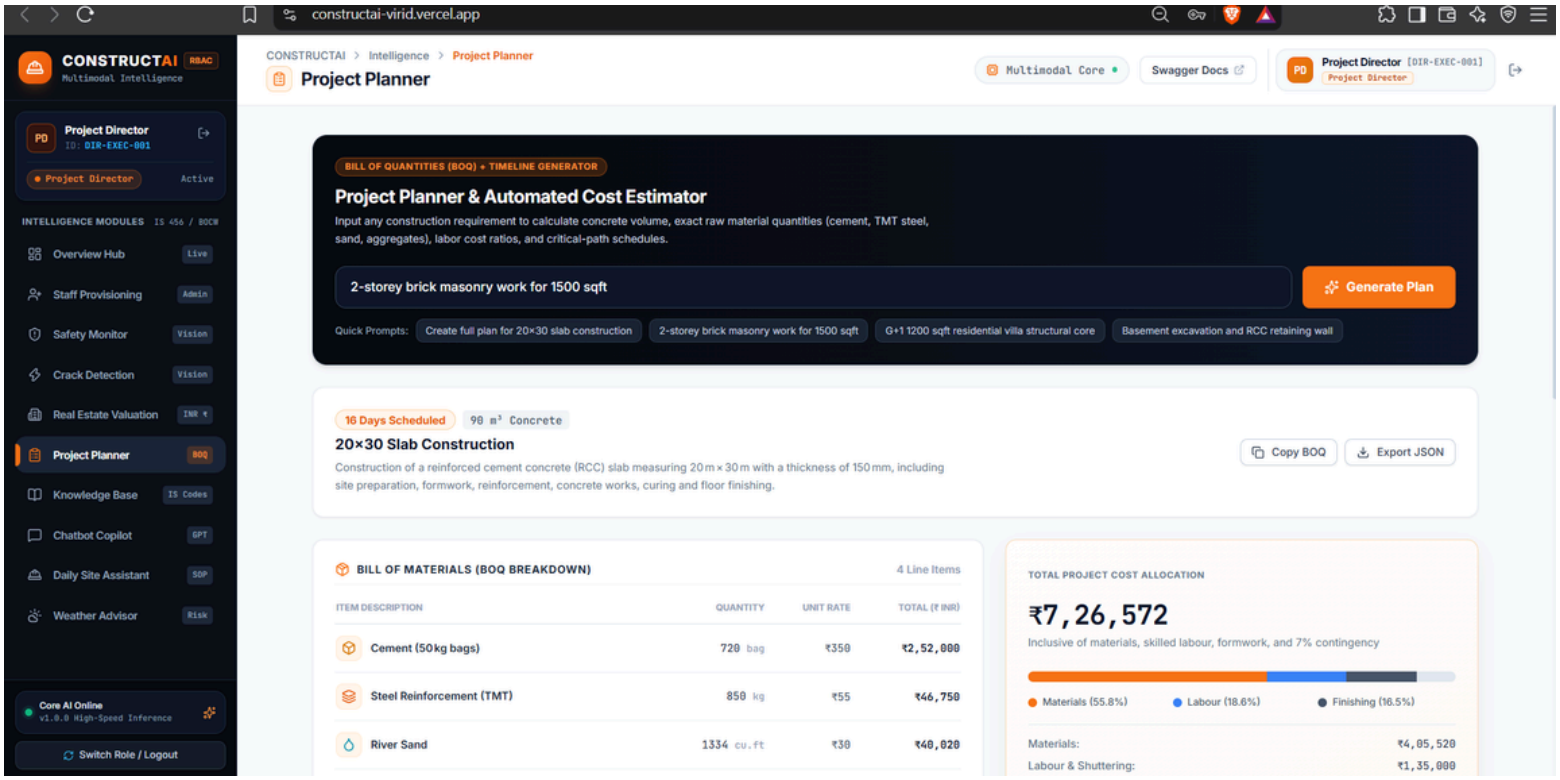
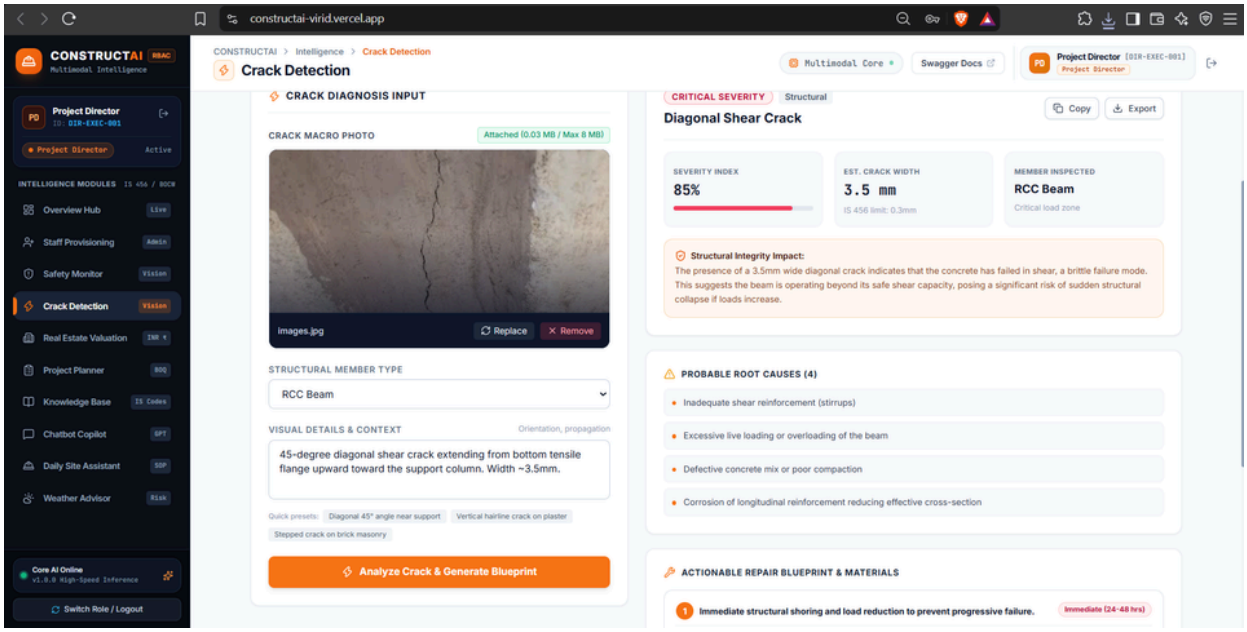


Figure 19.1: Project Planner calculating material schedules and estimated project expenditure.

Description: Accepts inputs such as "2-storey brick masonry work for 1500 sqft" and outputs estimated material quantities (cement bags, steel tonnage, sand volume), financial projections (₹7,26,572 estimated cost), and a phased critical-path timeline.

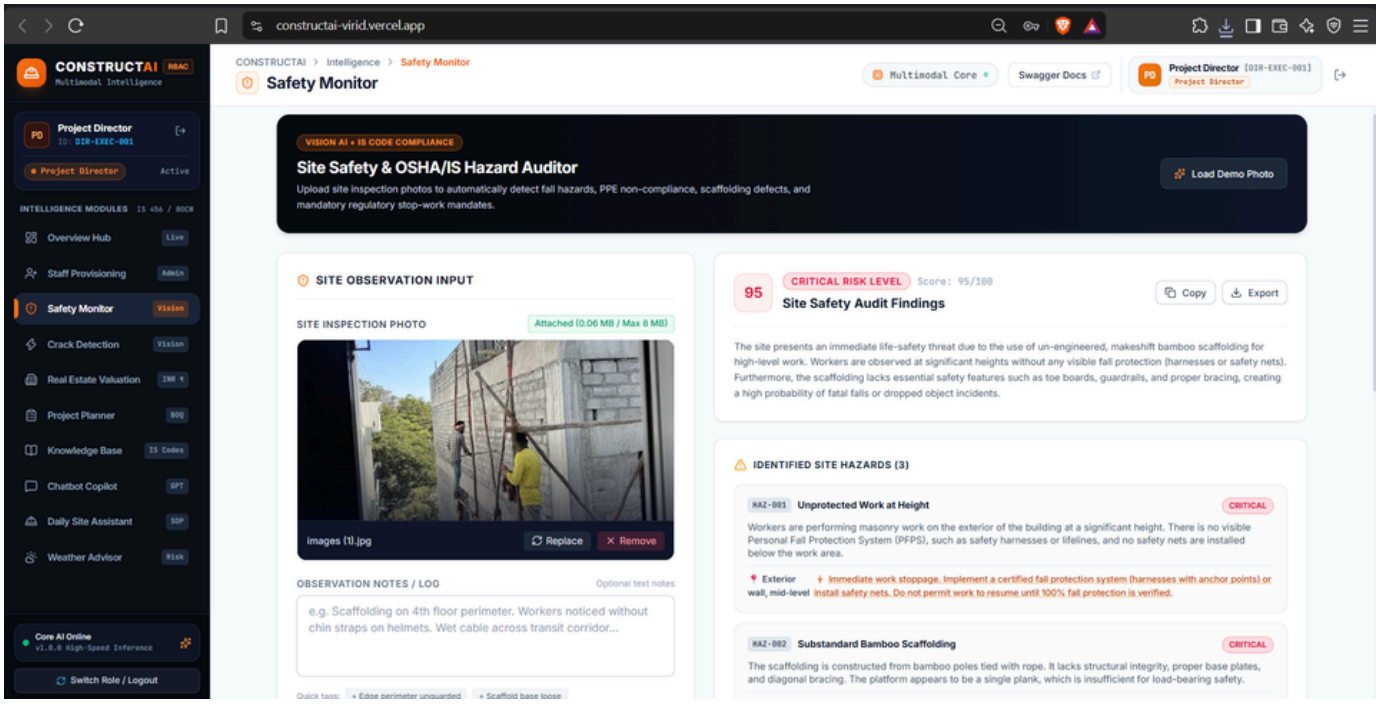
Web Application Exploration: Diagnostics & Compliance

Structural Concrete Crack Detection Module



Crack Diagnostic Interface: Evaluates uploaded concrete images against IS 456:2000 serviceability limits. Identifies a 3.5 mm shear crack, flags an 85% severity index, and prescribes epoxy pressure grouting and carbon fiber wraps.

Site Safety & BOCW 1996 Hazard Auditor



Safety Audit Interface: Inspects active site photos for OSHA and BOCW compliance. Identifies missing PPE and defective bamboo scaffolding, generates a 95/100 risk score, and issues a formal stop-work notice.

Conclusion & Future Scope

ConstructAI demonstrates the practical value of applying multimodal foundation models to the civil engineering and construction industry. By combining Groq LPU hardware acceleration with strict civil engineering codices—specifically IS 456:2000 for concrete limit-state forensics and BOCW 1996 for job-site safety auditing—the platform delivers accurate, sub-second diagnostics that bridge traditional field methods with automated visual intelligence.

The complete 21-page system architecture reflects the six foundational milestones established by SmartBridge and Skill Wallet, spanning initial environment setup and Groq API provisioning through to asynchronous FastAPI development, responsive React interface design, and end-to-end edge-case testing.

Future Roadmap:

- **BIM Model Integration (IFC Parsing):** Expanding the planning pipeline to import Industry Foundation Classes (.ifc) files for automated 3D quantity take-offs.
- **Edge AI Deployment for Drones:** Packaging optimized vision weights onto embedded platforms (NVIDIA Jetson) for real-time aerial safety audits.
- **Automated IoT Sensor Telemetry:** Ingesting real-time sensor streams from concrete maturity meters to provide accurate curing predictions based on ambient weather.